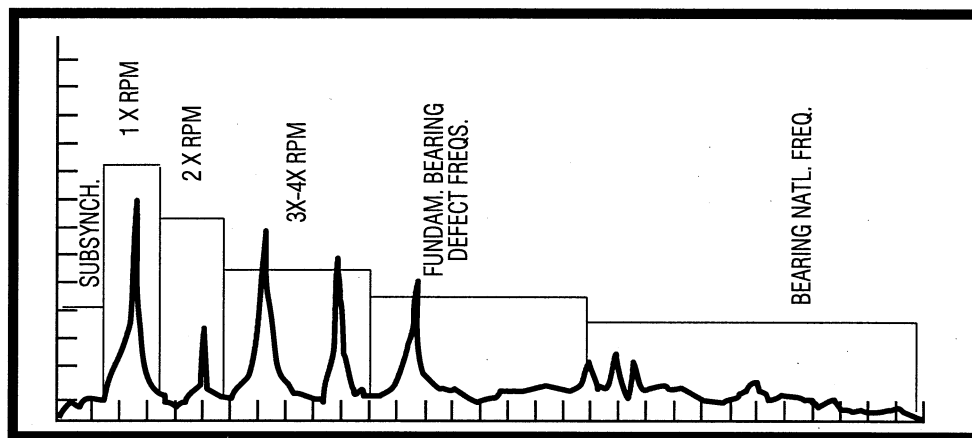




ANALYSIS II

CONCENTRATED VIBRATION SIGNATURE ANALYSIS AND RELATED CONDITION MONITORING TECHNIQUES



Authored By:

Mr. James E. Berry P.E.
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SPECIALISTS IN PREDICTIVE MAINTENANCE, MACHINERY DIAGNOSTICS, AND VIBRATION REDUCTION

PREDICTIVE MAINTENANCE AND VIBRATION SIGNATURE ANALYSIS II

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**SPECIALISTS IN PREDICTIVE MAINTENANCE, MACHINERY DIAGNOSTICS,
AND VIBRATION REDUCTION**

We thank you for the opportunity to assist you in development of your vibration analysis and condition monitoring skills. As you can see from the length and breadth of the information within this text, many variables go into application of the material and so much depends upon the knowledge and judgement of the analyst and/or technician. The author and his company and /or companies can assume no responsibility for results in spite of many years of successful and trouble-free personal application.

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ANALYSIS II

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FIBERGLASS _____	HVAC _____	INSTRUM./SWARE _____
MINING _____	PETROLEUM _____	PHARMACEUTICAL _____
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TEXTILES _____	TOBACCO _____	UTILITIES _____
VIB. SERVICES _____	TRAINING SERVICES _____	OTHER (SPECIFY) _____

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 DO YOU HAVE AN ONGOING VIBRATION/PdM PROGRAM? YES ✓ NO _____
 IF YES, HOW LONG HAS PROGRAM BEEN IN PLACE? 7 YRS.
 PERSON RESPONSIBLE FOR PROGRAM? NAME: Tom Smith
 TITLE: Reliability Engineer

C. WHAT TYPE OF VIBRATION INSTRUMENTS ARE USED AT YOUR PLANT?

(ENTER BOTH MANUFACTURER MODEL NO. AND QUANTITY: i.e., CSI 2130 (3))

• OVERALL LEVEL METER <u>SKF PenPlus(3)</u> (i.e., SKF Pen Plus)	• SWEPT-FILTER ANALYZER <u>IRP 880(1)</u> (i.e., SCHENCK VIBRAPORT 30)
• DATA COLLECTOR <u>SKF CMVA 60(2)</u> (i.e., Entek DataPAC 1500)	• 2 CHANNEL RTA _____ (i.e., HP 35670)
• MULTI-CHANNEL DATA <u>CSI 2130(4)</u> COLLECTOR (i.e., CSI 2130)	• MULTI-CHANNEL RTA <u>Iotech ZonicBook</u> (i.e., SA 395)

D. WHAT IS THE BRAND NAME AND MODEL # OF THE PRIMARY DATA COLLECTOR USED AT YOUR PLANT? (INDICATE MODEL NO. IN BLANK: i.e., SKF CMVA-60)

BENTLY _____	COMMTTEST _____	CSI <u>2130</u>	DI _____	DLI _____
ENTEK IRD _____	PRUFTECHNIK _____	SKF _____	OTHER _____	

E. WHAT PMP SOFTWARE AND VERSION # DO YOU USE? (INDICATE ALL SOFTWARE THAT YOU NOW USE)

ALERT _____	ASCENT _____	IQ 2000 _____	MACHINE ANALYST <u>V 2.2</u>
MASTERTREND _____	ODYSSEY _____	OMNITREND _____	PRISM ⁴ _____
RBMWARE <u>v. 4.81</u>	TRENDMASTER _____	OTHER _____	

F. EXPERT DIAGNOSTIC SOFTWARE USED AT YOUR PLANT: (PLACE CHECK MARK)

EXPERTALERT _____ MDU _____ MOTORCHECK _____
 MOTORMONITOR _____ NSPECTR II ☒ PRISM4 PRO _____ ROMEX _____
 OTHER (PLEASE IDENTIFY) _____ NONE

G. EXPERIMENTAL MODAL ANALYSIS SOFTWARE USED AT YOUR PLANT (CHECK ALL THOSE WHICH YOU NOW USE).

LMS _____ ME'scope ☒ SDRC _____ STAR _____ ZONIC _____ OTHER (SPECIFY) _____

H. HAVE YOU PERFORMED FINITE ELEMENT MODELING? IF SO, WITH WHAT SOFTWARE?

YES _____ NO ☒ SOFTWARE USED? _____

WHAT OTHER VIBRATION/PREDICTIVE MAINTENANCE SEMINARS HAVE YOU ATTENDED?

SEMINAR NAME	SEMINAR COMPANY	APPROX. DATE ATTENDED
Entry Level	Technical Associates	Feb., 2001
Dynamic Balancing	Technical Associates	June, 2001
Analysis I	Technical Associates	June, 2003

J. IS THERE ANY SPECIAL MACHINE TYPE IN YOUR FACILITY ON WHICH YOU NEED HELP? AND/OR WOULD LIKE ADDITIONAL INFORMATION?

MACHINE NAME	BEARING TYPE	RPM	HP	DRIVE TYPE
Hoffman Blower (7-stage)	R.E. Bearing	3580	150 HP	Direct Coupled
Buffalo Forge Scrubber Fan	R.E. Bearing	1180	100 HP	Belt Drive

K. PLEASE LIST ANY SPECIAL QUESTIONS OR TOPICS THAT YOU WOULD LIKE COVERED DURING THIS SEMINAR?

1) How can you reliably diagnose problems with Rolling Element Bearings?
 2) Is there now any technique available to detect problems on an intermediate gear inside a 4-stage Gearbox?
 3) What is PeakVue and how should I set it up for field measurements?

L. HOW DID YOU FIND OUT ABOUT THIS SEMINAR? (PLEASE CHECK PRIMARY CONTACT)

BROCHURE ☒ DIRECT SALES _____ PHONE CALL _____
 COLLEAGUE _____ FORMER ATTENDER ☒ TA WEB SITE ☒
 MAGAZINE AD (If So, Which Magazine)? _____ OTHER (Please Indicate Source) _____

M. WHAT PROMPTED YOU TO ATTEND THIS PARTICULAR SEMINAR? (PLEASE CHECK ALL THAT APPLY.)

ADVANCEMENT ☒ SEMINAR LOCATION _____ SPEAKER _____
 SUBJECT MATTER _____ WAS RECOMMENDED ☒ OTHER _____

PLEASE SUBMIT THIS COMPLETED FORM AND MAIL/FAX TO TECHNICAL ASSOCIATES BEFORE ATTENDING SEMINAR, THANK YOU. WE APPRECIATE YOUR TAKING THE TIME TO HELP US CUSTOMIZE AND ENHANCE THIS SEMINAR TO ENSURE WE MEET PARTICULAR NEEDS.

TECHNICAL ASSOCIATES OF CHARLOTTE, P.C.

TECHNICAL ASSOCIATES PRE-SEMINAR QUESTIONNAIRE

YOUR NAME: _____ SEMINAR DATES: _____
TITLE: _____ SEMINAR LOCATION: _____
COMPANY: _____ SEMINAR LEVEL (CIRCLE) BAL ENTRY I II III ADV MODAL
LOCATION: _____ VIBRATION EXPERIENCE: YRS. _____ MOS. _____
PHONE: _____ EXT: _____ SPECTRAL ANAL. EXPER: YRS. _____ MOS. _____
EMAIL: _____ FAX: _____

PLEASE COMPLETE THE FOLLOWING QUESTIONNAIRE, THIS INFORMATION IS FOR USE IN THIS CLASS AND TO HELP US PLAN FUTURE SEMINARS: (SEE SAMPLE PRE-SEMINAR SURVEY ENCLOSED)

A. TYPE OF INDUSTRY AND PRODUCT MANUFACTURED/SERVICES RENDERED BY YOUR COMPANY:

AUTOMOTIVE _____	CHEMICAL _____	CONSUMER PRODS. _____
CONSULTANT _____	FOOD/BEVERAGES _____	FABRICA. METAL _____
FIBERGLASS _____	HVAC _____	INSTRUM./S'WARE _____
MINING _____	PETROLEUM _____	PHARMACEUTICAL _____
PULP & PAPER _____	REPAIR _____	SYNTHETIC FIBER _____
TEXTILES _____	TOBACCO _____	UTILITIES _____
VIB. SERVICES _____	TRAINING SERVICES _____	OTHER (SPECIFY) _____

B. COMPANY INFORMATION:

PRIMARY PRODUCT(S) AT YOUR FACILITY? _____
NUMBER OF EMPLOYEES AT YOUR FACILITY? _____
DO YOU HAVE AN ONGOING VIBRATION/PdM PROGRAM? YES _____ NO _____
IF YES, HOW LONG HAS PROGRAM BEEN IN PLACE? _____ YRS.
PERSON RESPONSIBLE FOR PROGRAM? NAME: _____
TITLE: _____

C. WHAT TYPE OF VIBRATION INSTRUMENTS ARE USED AT YOUR PLANT? (ENTER BOTH MANUFACTURER MODEL NO. AND QUANTITY: i.e., CSI 2130 (3))

• OVERALL LEVEL METER _____ (i.e., SKF Pen Plus)	• SWEPT-FILTER ANALYZER _____ (i.e., SCHENCK VIBRAPORT 30)
• DATA COLLECTOR _____ (i.e., Entek DataPAC 1500)	• 2 CHANNEL RTA _____ (i.e., HP 35670)
• MULTI-CHANNEL DATA _____ COLLECTOR (i.e., CSI 2130)	• MULTI-CHANNEL RTA _____ (i.e., SA 395)

D. WHAT IS THE BRAND NAME AND MODEL # OF THE PRIMARY DATA COLLECTOR USED AT YOUR PLANT? (INDICATE MODEL NO. IN BLANK: i.e., SKF CMVA-60)

BENTLY _____ COMMTTEST _____ CSI _____ DI _____ DLI _____
ENTEK _____ PRUFTECHNIK _____ SKF _____ OTHER _____

E. WHAT PMP SOFTWARE AND VERSION # DO YOU USE? (INDICATE ALL SOFTWARE THAT YOU NOW USE)

ALERT _____	ASCENT _____	IQ 2000 _____	MACHINE ANALYST _____
MASTERTREND _____	ODYSSEY _____	OMNITREND _____	PRISM ⁴ _____
RBWARE _____	TRENDMASTER _____		OTHER _____

F. EXPERT DIAGNOSTIC SOFTWARE USED AT YOUR PLANT: (PLACE CHECK MARK)

EXPERTALERT _____ MDU _____ MOTORCHECK _____
MOTORMONITOR _____ NSPECTR II _____ PRISM4 PRO _____ ROMEX _____
OTHER (PLEASE IDENTIFY) _____ NONE _____

G. EXPERIMENTAL MODAL ANALYSIS SOFTWARE USED AT YOUR PLANT (CHECK ALL THOSE WHICH YOU NOW USE).

LMS _____ ME'scope _____ SDRC _____ STAR _____ ZONIC _____ OTHER (SPECIFY) _____

H. HAVE YOU PERFORMED FINITE ELEMENT MODELING? IF SO, WITH WHAT SOFTWARE?

YES _____ NO _____ SOFTWARE USED? _____

I. WHAT OTHER VIBRATION/PREDICTIVE MAINTENANCE SEMINARS HAVE YOU ATTENDED? SEMINAR NAME SEMINAR COMPANY APPROX. DATE ATTENDED

J. IS THERE ANY SPECIAL MACHINE TYPE IN YOUR FACILITY ON WHICH YOU NEED HELP? AND/OR WOULD LIKE ADDITIONAL INFORMATION?

MACHINE NAME BEARING TYPE RPM HP DRIVE TYPE

K. PLEASE LIST ANY SPECIAL QUESTIONS OR TOPICS THAT YOU WOULD LIKE COVERED DURING THIS SEMINAR?

L. HOW DID YOU FIND OUT ABOUT THIS SEMINAR? (PLEASE CHECK PRIMARY CONTACT)

BROCHURE _____ DIRECT SALES _____ PHONE CALL _____
COLLEAGUE _____ FORMER ATTENDER _____ TA WEB SITE _____
MAGAZINE AD (If So, Which Magazine)? _____ OTHER (Please Indicate Source) _____

M. WHAT PROMPTED YOU TO ATTEND THIS PARTICULAR SEMINAR? (PLEASE CHECK ALL THAT APPLY.)

ADVANCEMENT _____ SEMINAR LOCATION _____ SPEAKER _____
SUBJECT MATTER _____ WAS RECOMMENDED _____ OTHER _____

PLEASE SUBMIT THIS COMPLETED FORM AND MAIL/FAX TO TECHNICAL ASSOCIATES BEFORE ATTENDING SEMINAR, THANK YOU. WE APPRECIATE YOUR TAKING THE TIME TO HELP US CUSTOMIZE AND ENHANCE THIS SEMINAR TO ENSURE WE MEET PARTICULAR NEEDS.

TECHNICAL ASSOCIATES OF CHARLOTTE, P.C.

TECHNICAL ASSOCIATES OF CHARLOTTE, P.C.

POST-SEMINAR EVALUATION

SEMINAR DATES: _____ SEMINAR LEVEL (CIRCLE): ENTRY I II III ADV.
 ATTENDEE NAME: _____ TITLE: _____
 COMPANY NAME: _____ YOUR VIBRATION EXPERIENCE: YEARS _____ MONTHS _____
 COMPANY LOCATION: _____ YOUR SPECTRAL ANAL. EXPER: YEARS _____ MONTHS _____
 SEMINAR LOCATION: _____
 INSTRUCTOR: _____

EVALUATION TOPICS RATING >	LOW HIGH									
	1	2	3	4	5	6	7	8	9	10
1. EVALUATE CONTENT OF THIS SEMINAR.										
2. EVALUATE PERFORMANCE OF INSTRUCTOR:										
A. KNOWLEDGE OF SUBJECT										
B. PRESENTATION CAPABILITIES										
C. ENTHUSIASM DISPLAYED										
D. OVERALL PERFORMANCE RATING										
3. HOW APPLICABLE TO YOUR JOB HAS BEEN THE CONTENT OF THIS SEMINAR AND SEMINAR TEXT?										
4. CAN WHAT YOU HAVE LEARNED MAKE YOU MORE EFFECTIVE IN YOUR JOB?										
5. HOW WOULD YOU RATE THE SEMINAR FACILITIES:										
A. MEETING ROOM -										
B. SLEEPING ROOM -										
C. FOOD SERVICE -										
6. ARE YOU INTERESTED IN FURTHER TRAINING IN THIS AREA OF VIBRATION/PREDICTIVE MAINTENANCE?										

CONCENTRATION ON SEMINAR:

- ON WHAT SHOULD MORE TIME BE CONCENTRATED? _____

- ON WHAT SHOULD LESS TIME BE CONCENTRATED? _____

- WHAT WOULD YOU SUGGEST TO IMPROVE FUTURE SEMINARS? _____

- DO YOU HAVE ANY ADDITIONAL COMMENTS ABOUT THIS SEMINAR, THE INSTRUCTORS, THE COMPANY OR THE FACILITIES? _____

WE WANT TO GENUINELY THANK YOU FOR THE PRIVILEGE OF SHARING WITH YOU THIS WEEK.
 WE HOPE TO SEE YOU AGAIN IN THE FUTURE.

R-9707-1

RECOMMENDED PERIODICALS FOR THOSE INTERESTED IN PREDICTIVE MAINTENANCE

1. Sound and Vibration Magazine

P.O. Box 40416
Bay Village, OH 44140
Mr. Jack Mowry, Editor and Publisher
Phone: 216-835-0101
Fax : 216-835-9303

Terms: Normally free for bona fide qualified personnel concentrating in the Sound and Vibration Analysis/Plant Engineering Technologies. Non-qualified personnel - \$25/per year within the U.S.

Comments: This is a monthly publication that normally will include approximately 4-6 issues per year devoted to Predictive Maintenance. Their Predictive Maintenance articles are usually practical and in good depth; normally contain real "meat" for the PPM vibration analyst. Sound and Vibration has been published for over 25 years.

2. Vibrations Magazine

The Vibration Institute
6262 South Kingery Hwy, Suite 212
Willowbrook, IL 60514
Institute Director - Dr. Ronald Eshleman
Phone: 630-654-2254
Fax : 630-654-2271

Terms: Vibrations Magazine is sent to Vibration Institute members as part of their annual fee, (approx. \$45 per year). It is available for subscription to nonmembers at \$55/per year; \$60/foreign.

This is a quarterly publication of the Vibration Institute. Always contains very practical and useful Predictive Maintenance Articles and Case Histories. Well worth the small investment.

Comments: Yearly Vibration Institute fee includes reduced proceedings for that year if desired for the National Conference normally held in June. They normally meet once per year at a fee of about \$675/per person, (\$600/person for Institute members) including conference proceedings notes and mini-seminar papers. All of the papers presented, as well as mini-courses, at the meeting are filled with "meat" for the Predictive Maintenance Vibration Analyst. Vibrations Magazine was first published in 1985 although the Institute has been in existence since approximately 1972, with their first annual meeting in 1977. The Vibration Institute has several chapters located around the United States which normally meet on a quarterly basis. The Carolinas' Vibration Institute Chapter normally meets in Greenville, SC; Charleston, SC; Columbia, SC; Charlotte, NC; Raleigh, NC; and in the Winston Salem, NC areas. For Institute membership information, please contact: Dr. Ron Eshleman at 630-654-2254. When doing so, be sure to ask what regional chapter is located to your area. Membership fees for the "Annual Meeting Proceedings" are \$30/per year (normal cost is approx. \$60/per year for proceedings if annual meeting is not attended). Please tell Ron that we recommended you joining the Vibration Institute when you call or write to him.

R-0697-1

3. P/PM Technology Magazine

P.O. Box 1706
Minden, NV 89423-1706 (Pacific Coast Time)
Phone: 702-267-3970; 800-848-8324
Fax : 702-267-3941
Publisher- Mr. Ronald James; Assistant Publisher: Susan Estes

Terms: \$42/per year for qualified USA subscribers, (individuals and establishments involved with industrial plant and facilities maintenance; subscribers must be associated in engineering, maintenance, purchasing or management capacity). \$60/year for unqualified subscribers.

Comments: This is a bimonthly magazine with articles about all facets of PPM Technologies, including Vibration Analysis, Oil Analysis, Infrared Thermography, Ultrasonics, Steam Trap Monitoring, Motor Current Signature Analysis, etc. These are normally good practical articles. Also includes some cost savings information, although does not necessarily include how these cost savings were truly determined. P/PM Technology also hosts at least one major conference per year in various parts of the United States. Intensive training courses in a variety of condition monitoring technologies will also be offered in vibration analysis, root cause failure analysis, oil analysis, thermographic analysis, ultrasonic analysis, etc..)

4. Maintenance Technology Magazine

1209 Dundee Ave., Suite 8
Elgin, IL 60120
Phone: 800-554-7470
Fax : 804-304-8603
Publisher: Arthur L. Rice

Terms: \$95/per year for non-qualified people This is a monthly magazine that usually has at least one article relating to Predictive Maintenance using vibration analysis within each issue. In addition to vibration, it likewise always offers other articles covering the many other technologies now within Predictive Maintenance.

5. Reliability Magazine

PO Box 856
Monteagle, TN 37356
Phone: 423-592-4848
Fax : 423-592-4849

Editor: Mr. Joseph L. Petersen

Terms: \$49 per year in USA; \$73 per year outside USA.

Comments: This bimonthly magazine covers a wide variety of Condition Monitoring Technologies including Vibration Analysis, Training, Alignment, Infrared Thermography, Balancing, Lubrication Testing, CMMS and a unique category they entitle "Management Focus".

NOTE: In addition to these periodicals, many of the major predictive maintenance hardware and software vendors put out periodic newsletters. Some of these in fact do include some "real meat" in addition to their sales propaganda. We would recommend that you contact, particularly the vendor supplying your predictive maintenance system for their newsletter. Their newsletter will likewise advise you of updates in their current products.

R-0697-1

ANALYSIS I SEMINAR AGENDAS

INTRODUCTION TO VIBRATION ANALYSIS AND PREDICTIVE MAINTENANCE

(Recommended Full-time PMP Vibration Experience = 0 to 6 Months)

1. Seminar Overview
2. What is Vibration and How Can it be Used to Evaluate Machinery Condition?
 - What is Frequency and How Does it Relate to a Time Waveform?
 - What is Vibration Displacement, Velocity, and Acceleration?
 - How to Convert From One Vibration Parameter to Another (in/sec to g, etc.)
 - What is Phase?
 - What is a Vibration Spectrum (Also Called an "FFT" or "Signature")?
 - Effect on Frequency Accuracy of No. of FFT Lines
 - Difference Between RMS, Peak, and Peak-To-Peak Amplitude and its Significance
 - When to Use Displacement, Velocity, or Acceleration
 - How Much is Too Much Vibration?
3. Overview of the Strengths and Weaknesses of Typical Vibration Instruments
4. Overview of Various Vibration Transducers and Their Optimum Applications
5. Role of Spike Energy, High Frequency Detection (HFD), and Shock Pulse, along with Proper Specification of Their Alarm Levels at Various Speeds
6. Introduction to Vibration Signature Analysis and How it is Used to Evaluate Machine Operating Condition
Overview of 5-Page "Illustrated Vibration Diagnostic Chart"
 - Mass Unbalance
 - Eccentric Rotors
 - Bent Shaft
 - Misalignment
 - Mechanical Looseness, Improper Component Fit, and Soft Foot
 - Tracking of Rolling Element Bearing Health Using Vibration Signature Analysis
 - Belt Drive Problems
 - Electrical Problems in Motors
7. Proven Method for Specifying Spectral Band Alarm Levels and Frequencies Using Today's Predictive Maintenance Software Systems
8. Common Pitfalls in Everyday Vibration Measurements and Their Effect on Properly Detecting and Diagnosing Machine Problems
 - Proper and Improper Measurement Locations
 - Effect of Transducer Mounting on Resulting Vibration Spectra
 - Incorrect Transducer Choice
 - Incorrect Measurement Parameter (mils, in/sec, g, gSE, etc.)
 - Incorrect Frequency Span
 - Poor Measurement Techniques
9. Setup and Implementation of Effective Predictive Maintenance and Condition Monitoring Programs
10. Actual Case Histories of Vibration Diagnostics on Various Machine Types

ANALYSIS II SEMINAR AGENDAS

CONCENTRATED VIBRATION SIGNATURE ANALYSIS AND RELATED CONDITION MONITORING TECHNIQUES

(Recommended Full-time PMP Vibration Experience = 6 to 12 Months)

1. Seminar Overview
2. Brief Review of Analysis I Seminar Topics
3. Proper Use of Vibration Severity Charts
4. How to Refine Overall Alarm and Spectral Band Alarm Levels after Several surveys have been taken
5. Phase Analysis Techniques
6. Detailed Vibration Signature Analysis for 40 Machine Problems Using Technical Associates' 5-Page **"Illustrated Vibration Diagnostics Chart"**:
 - Mass Unbalance
 - Eccentric Rotors
 - Bent Shaft
 - Misalignment and Coupling Problems
 - Machinery Failures Due to Resonant Vibration
 - Mechanical Looseness - 3 Distinctly Different Types
 - Rotor Rub
 - Sleeve Bearing Problems
 - Rolling Element Bearing Problems
 - Flow-Induced Vibration (Cavitation, Starvation, etc..)
 - Gear Problems
 - Electrical Problems in Motors
 - Belt-Drive Problems
 - Beat Vibration Problems
 - Soft Foot and Distorted Frame Problems
7. Presentation of a Proven Analysis Procedure Using the **"5-Page Illustrated Vibration Diagnostic Chart"** (Including Examples)
8. What is "High Frequency Enveloping and Demodulation Spectral Analysis" (Used to Detect Problems with Rolling Element Bearings, Gears, Lubrication, etc.)?
9. How to Select the Optimum Condition Monitoring Hardware and Software Using Proven Weighted Matrix Methods Based on Cost, Hardware Capabilities, Software Capabilities and Vendor History (Generic Procedure Independent of PMP Vendor)
10. Presentation of **Real-World Case Histories**

ANALYSIS III SEMINAR AGENDAS

INTRODUCTION TO SPECIAL VIBRATION ANALYSIS TECHNIQUES AND NARROWBAND SPECTRAL ENVELOPE ALARMS

(Recommended Full-time PMP Vibration Experience = 12 to 24 Months)

1. Seminar Overview
2. Review of Analysis II Seminar Topics
3. Introduction to Frequency and Amplitude Interpolation and Why They are so Important to Accurate Vibration Signature Analysis
4. How Can One Diagnose a Bearing, Gear or Electrical Problem When He Does Not Know Rolling Element Bearing Model Numbers, Gear Tooth Count, or Number of Rotor Bars?
5. How to Properly Create and Specify Optimum Narrowband Envelope Spectral Alarms for Various "Families" of Machines Having Different Operating Speeds
6. How Much is Too Much at Specific Vibration Frequencies for Various Machine Types?
 - At Operating Speed (1x RPM)
 - At 2x RPM
 - At 3x RPM
 - At 2x Line Frequency in an Induction Motor
 - At 1x and 2x Rotor Bar Pass Frequency in an Induction Motor
 - At Blade Pass Frequency and Harmonics
 - At Gear Mesh Frequency and Harmonics
 - At Belt Frequencies and Harmonics
 - At Rolling Element Bearing Frequencies and Harmonics
 - At SCR Firing Frequencies in DC Motors
 - At Other Electrical Line Frequencies in DC Motors
 - At Coil Passing Frequency in Synchronous Motors
7. Acceptance Tests and Criteria for New and/or Overhauled Machinery
8. How to Accurately Track Rolling Element Bearing Condition Using Both Vibration Signature Analysis and High Frequency Enveloping and Demodulated Spectral Techniques
9. Introduction to Gear Problem Diagnostics
10. Natural Frequency Testing
 - Impulse Natural Frequency Tests
 - Coastdown and Runup Tests
 - Bode and Nyquist Plotting (Phase and Amplitude vs. RPM)
 - Swept-sine Variable Frequency Shakers
11. Introduction to Special Diagnostic Techniques
 - Low Frequency Vibration Measurement Considerations (Excerpts from "Advanced 1" Seminar Text)
 - Operating Deflected Shape Analysis (ODS) by Manual as well as by Automated Software-driven Methods
 - Time Waveform Analysis and how it Relates to FFT Analysis
 - Synchronous Time Averaging (Excerpts from "Advanced I" Seminar Text)
 - Introduction to Spectral Analysis of Motor Current

12. Presentation of Effective PMP Reporting Techniques to Keep Program Successes Visible and to Ensure Proper Corrective Actions are Taken
13. Considerations for Vibration Analysis of Specific Machine Types:
 - Horizontal Pumps
 - Vertical Pumps
 - Centrifugal Air Compressors
 - Centrifugal Fans
14. Presentation of **Real-World Case Histories**

ADVANCED IV SEMINAR AGENDAS

ADVANCED VIBRATION DIAGNOSTIC TECHNIQUES AND HOW TO PROPERLY EVALUATE LOW, HIGH, AND VARIABLE SPEED MACHINES

(Recommended Full-Time PMP Vibration Experience = 24 to 48 Months)

1. Seminar Overview
2. Review of Analysis III Seminar Topics
3. More on the High Frequency Enveloping and Demodulation Techniques
 - What it Means
 - Setting Up the Parameters (Filters, Frequency Spans, etc.)
 - How to Interpret the Results
 - Examples of High Frequency Enveloping Spectra ,
4. Special Analysis Techniques
 - Time Waveform Analysis - When and Where it Detects Special Problems FFT Signature Analysis Misses
 - Synchronous Time Averaging - How it is Used to Pinpoint Problems Other Techniques Cannot
 - Order Tracking - Why it is Especially Important to Properly Evaluate Variable-Speed Machines
 - Motor Current Analysis and Presentation of a "Motor Current Troubleshooting and Severity Chart" Developed by Technical Associates
 - Advanced Natural Frequency Testing and Confirmation of What Component is Resonant
 - Introduction to Experimental Modal Analysis Using Multi-channel Analyzers
 - Use of Real-Time Analyzers (Single and Multi-Channel)
5. Low-Speed Machine Vibration Analysis Techniques and Instrumentation Requirements (Predominantly Below 100 RPM)
6. High-Speed Machine Vibration Analysis Techniques, Mounting and Instrumentation Requirements (Generating Frequencies Exceeding 10,000 Hz or 600,000 CPM)
7. How to Analyze Variable Speed Machines Both on a One-Time Diagnostic Basis as well as in a Condition Monitoring Program
8. Gaining a Better Understanding of What Today's Vibration Analyzers are Doing in Practical Terms
 - Significance of Dynamic Range and Number of A/D Bits
 - Significance of Measurement Bandwidth (i.e., "How Far Apart do Two Peaks Have to be for an FFT Analyzer to Display Two Separate Frequencies"?)
 - What are the Real Advantages and Disadvantages of Using Various FFT Windows (i.e., Hanning, Flattop, Uniform, etc.)?
 - Overlap Processing - What is it, Along with its Advantages and Disadvantages?
 - Interpolated Frequency Enhancement - What is it and Why is it so Significant to Accurate Machinery Problem Diagnostics?
9. Presentation of Real-World Case Histories

